



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

Fax: +66-2721-3515

SAFETY DATA SHEET

Company Identification

Thep Watana Co., Ltd. Address: 382 Bangpoo Industrial Estate.

Sukhumvit Road, Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66(0)-2721-3510;

Fax: +66(0)-2721-3515

1. CHEMICAL IDENTIFICATION

Product Identification

Product name: butachlor 60% W/V EC

Product use: Herbicide

Relevant identified uses of the substance/mixture and uses advised against:

Used as a selective pre-emergent herbicide

2. HAZARD IDENTIFICATION

Label Precautionary Statements

Harmful

Harmful If Swallowed.

Do Not Breathe Vapor.

Avoid Contact With Skin And Eyes.

3. COMPOSITION/INFORMATION ON INGREDIENT

Chemical name: *N*-(Butoxymethyl)-2-chloro-*N*-(2,6-diethylphenyl)acetamide (IUPAC)

Common name: butachlor

Synonyms: Acetamide, *N*-(Butoxymethyl)-2-Chloro-*N*-(2,6-Diethylphenyl)- (9ci) *

Acetanilide, *N*-(Butoxymethyl)-2-Chloro-2',6'-Diethyl- (8ci) *

Aimchlor * Bilchlor * Butachlor * Butaclor * Butanex *

N-(Butoxymethyl)-2-Chloro-2',6'-Diethylacetanilide *

N-(Butoxymethyl)-2-Chloro-*N*-(2,6-Diethylphenyl)Acetamide *

2-Chloro-2',6'-Diethyl-*N*-(Butoxymethyl)Acetanilide *

Cp 53619 * Delchlor * Delchlor 5g * Hiltachlor * Machete * Machete

(Herbicide) * Machette * Pillarsete * Rasayanchlor



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CAS Number: 23184-66-9

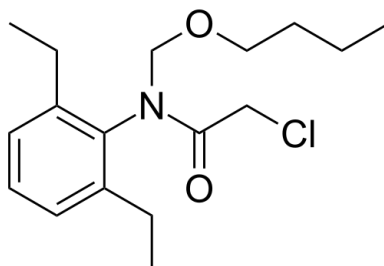
EC Number: 245-477-8

Ingredients and composition: Active ingredient – butachlor 60% W/V

Formula: $C_{17}H_{26}ClNO_2$

Molecular weight: 311.85 g/mol

Structural formula:



4. FIRST AID MEASURES

Description of first aid measures

In case of swallowed:

Wash out mouth with water provided person is conscious. Call a physician.

In case of eyes contact:

Flush with copious amounts of water for at least 15 minutes. Assure adequate flushing by separating the eyelids with fingers. Call a physician.

In case of skin contact:

Flush with copious amounts of water for at least 15 minutes. Remove contaminated clothing and shoes. Call a physician.

In case of Inhaled contact:

Remove to fresh air. If not breathing give artificial respiration.

If breathing is difficult, give oxygen.

5. FIRE FIGHTING MEASURES

Flashpoint: > 60 °C

Flammability: Not flammable

Extinguishing Media:

Carbon dioxide, dry chemical powder or appropriate foam.

Water spray.



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Special Fire Fighting Procedures:

Wear self-contained breathing apparatus and protective clothing to

Prevent contact with skin and eyes.

Unusual fire and explosions hazards

Emits toxic fumes under fire conditions.

6. ACCIDENTAL RELEASE MEASURES (SPILLAGE)

Procedure for spill:

Wear self-contained breathing apparatus, rubber boots and heavy Rubber gloves.

Absorb on sand or vermiculite and place in closed containers for Disposal.

Ventilate area and wash spill site after material pickup is complete.

Evacuate area.

7. HANDLING AND STORAGE

Safe Handling:

Keep out of reach of children. Avoid contact with the eyes, skin and clothing and avoid inhalation of product or spray mist. If in eyes, wash it immediately with water. After handling and before eating, drinking or smoking, wash hands, arms and face thoroughly with soap and water. Do not contaminate food or use on household tanks.

Storage Conditions:

Store in the closed, original container in a dry, cool, well-ventilated area, secure area at minimum storage temperatures. Store in locked room or place away from children, animals, food, animal feed, seed and fertilizers.

For more detail, please refer to section 8.

8. EXPOSURE CONTROL/PERSONAL PROTECTION

Exposure standards

Safety shower and eye bath.

Mechanical exhaust required.

Avoid prolonged or repeated exposure.

Do not breathe vapor.

Avoid contact with eyes, skin and clothing.

Niosh/msha-approved respirator.

Compatible chemical-resistant gloves.

Chemical safety goggles.



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Store in a cool dry place.

Keep tightly closed.

Wash thoroughly after handling.

Personal Protective Equipment

Eyes Protection:

Wear OSHA-approved safety glasses with side-shields, safety goggles, or a full-face shield.

Skin Protection:

Wear chemical-resistant gloves such as Barrier Laminate, Neoprene Rubber, Nitrile Rubber, or Viton, and wear protective clothing.

Respiratory Protection:

Applicators using hand-held foggers in an enclosed area must wear a half-face, full-face, or hoodstyle NIOSH-approved respirator with a dust/ mist filtering cartridge (MSHA/ NIOSH approval number prefix TC21C), or a canister approved for pesticides (MSHA/ NIOSH approval number prefix TC-14G), or a cartridge or canister with any R, P or HE filter.

Protective clothing:

Wear chemical-resistant gloves, shoes and socks, long pants, and a long-sleeved shirt.

Work hygienic practices:

DO NOT SMOKE, EAT, OR DRINK, OR APPLY COSMETICS IN WORK AREA! Wash promptly if skin becomes contaminated. Wash at the end of each work shift and before eating, smoking, or using the toilet.

Other use precautions:

Use this product only in accordance with its labeling and with the Worker Protection Standard, 40 CFR, and Part 170. This Standard contains requirements for the protection of agricultural workers on farms, forests, nurseries, and greenhouses, and handlers of agricultural pesticides. It contains requirements for training, decontamination, notification, and emergency assistance. It also contains specific instructions and exceptions pertaining to the statements on the label about personal protective equipment (PPE), and restricted-entry interval. The requirements listed below only apply to uses of this product that are covered by the Worker Protection Standard.

Do not enter or allow worker entry into treated areas during the restricted entry interval (REI) of 12 hours.

PPE required for early entry to treated areas that is permitted under Worker Protection Standard and that involves contact with anything that has been treated, such as plants, soil, or water, is:

Coveralls;

Chemical-resistant gloves, such as Barrier Laminate, Nitrile Rubber, Neoprene Rubber, or Viton;

Shoes plus socks; Protective eyewear.



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Comments: Non-agricultural use requirements

The requirements in this section apply to uses of this product that are NOT within the scope of the Worker Protection Standard for agricultural pesticides (40 CFR, Part 170). The WPS applies when this product is used to treat agricultural plants on farms, forests, nurseries or greenhouses.

Keep unprotected persons out of treated areas until sprays have dried.

9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Physical stage:	Liquid
Color:	Clear, amber-brown color
Odor:	Sweet aromatic odor
pH:	5.0 – 8.0 at 20 °C
Initial boiling point:	156 °C
Melting/ Freezing Point:	-2.8 °C
Relative density:	1.07 g/cm ³ approx.
Solubility in water:	slightly soluble in water (emulsifiable)
Partition coefficient:	log Pow = 4.5 (n-octanol/water)

10. STABILITY AND REACTIVITY

Stability

Stable.

Incompatibilities

Strong Oxidizing Agents

Hazardous Combustion or Decomposition Products

Carbon Monoxide, Carbon Dioxide, Nitrogen Oxides, Hydrogen Chloride Gas.

Hazardous Polymerization

Will Not Occur.

11. TOXICOLOGICAL INFORMATION

Information on Acute toxicity for end product, butachlor:

Acute effects: ingestion of this product is harmful. Even small amounts of product may cause serious health problems (stomach pain, nausea, sickness, diarrhoea).

LD50 (Oral, rat): 1,740 mg/kg

LD50 (Dermal, rabbit): 1,740 mg/kg

LC50 (Inhalation, rat, 4h): > 5,300 mg/kg



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12. ECOLOGICAL INFORMATION

Toxicity

Adopt good working practices, so that the product is not released into the environment.

Data referred to the mixture, butachlor:

Fish – Acute toxicity:

Rainbow trout,

LC50 (96 h): 0.52 mg/L

Bluegill sunfish,

LC50 (96 h): 0.44 mg/L

Fathead minnow,

LC50 (96 h): 0.31 mg/L

Channel catfish,

LC50 (96 h): 0.10 – 0.14 mg/L

Crap,

LC50 (96 h): 0.62 mg/L

Daphnia magna (a small freshwater crustacean),

LC50 (48 h): 2.4 mg/L

Crayfish,

LC50 (96 h): 26 mg/L

Birds – Acute toxicity:

Bobwhite quail,

LC50 (5 days): 6,597 mg/kg (dietary)

Mallard ducks,

LD50: 4,640 mg/kg

LC50 (5 days): > 10,000 mg/kg (dietary)

Honey bee – Acute toxicity:

Highly toxic to bees

LD50 (contact): > 100 ug/bee

13. DISPOSAL CONSIDERATIONS

Dispose of empty, used containers by:

Contact a licensed professional waste disposal service to dispose of this material.

Dissolve or mix the material with a combustible solvent and burn in a Chemical incinerator equipped with an afterburner and scrubber.

Observe all federal, state and local environmental regulations.



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14. TRANSPORT INFORMATION

The data provided in this section is for information only. Please apply the appropriate regulations to properly classify your shipment for transportation.

UN No:	1992
Packaging Group:	II
Dangerous Goods Class:	3
Secondary risk:	6.1
Marine pollutant:	Yes

15. REGULATORY INFORMATION

European information

Risk Phrases:

R22: Harmful if swallowed.

Safety Phrases:

S23: Do not breathe vapor.

S24/25: Avoid contact with skin and eyes.

16. OTHER INFORMATION

The information given here is not necessarily exhaustive but is representative of relevant, reliable data.

Follow all local/regional/national/international regulations.

Please consult supplier if further information is needed.

Please read all labels carefully before using product.

This SDS summarizes our best knowledge of the health and safety hazard information of the product and how to safely handle and use the product in the workplace. Each user should read this SDS and consider the information in the context of how the product will be handled and used in the workplace including in conjunction with other products. If clarification or further information is needed to ensure that an appropriate risk assessment can be made, the user should contact this company. The responsibility for products sold is subject to our standard terms and conditions, a copy of which is sent to our customers and is also available on request.